Why light flickers for some but not all: an investigation into scintillation and angular size

Since eternity have I believed that stars twinkle because of their light. But a few minutes of thought would allow anyone, of any respectable age, to realise that this isn't the case. The sun, the star we know most of, doesn't twinkle. Any person who wouldn't know any better would assume, then, that distance is what makes stars twinkle. Let's assume this to be false for the time being, without any further consideration: we will return to this. Having said that, allow me to give a personal anecdote: while stargazing in a fairly remote location, I found a star in the winter night sky. Or at least that's what it seemed. It shone brighter than any other, and didn't twinkle. A quick search led me to believe it was Jupiter I was looking at. That begs the question: why wasn't Jupiter twinkling? Now, there *was* a slight twinkle, but on a clear night sky you'd find that it's a minimal movement in comparison to the other, seemingly smaller (or further) stars. My train of thought led me to reflect upon the size of the astral object… but obviously, hundreds of stars are visible from our earth as well, even astronomically larger than Jupiter, which we are using for our example. You would assume, then, that it was because of the reflection of the light from the sun reflecting on Jupiter's surface and bouncing back to us, and not its distance. Therefore, my second, incorrect, theory was: any light being reflected upon a surface cannot twinkle or, more correctly, scintillate. For our untrained eyes, we can easily say based on personal experience that this is the case. Light reflected from a mirror, for example, doesn't twinkle. This is false, and let us prove this together: There are countless instances of stars and surfaces in space twinkling, and perhaps looking out into the night sky would prove this! Satellites, although are much closer than any planet or star from us, in certain cases with the right (or wrong, in this specific case) conditions could reflect the light of the sun and scintillate. Assuming that what we just said is correct, the question is, does this apply to the astronomical scales we're referring to? Let's look at the moon. It reflects light back to our Earth, yes, but we know it doesn't scintillate. If we were to study an even further reflecting object, maybe at the same distance of a star, (assuming the surface reflecting the light is big enough to be seen from our earth) we could see a similar scintillation. But cause *doesn't* equal correlation.. And reflection is not a variable that changes the outcome, as we have verified. So we will keep studying this phenomenon, now also taking into consideration surfaces being reflected with light. My next theory now stemmed from this question: What do reflected light and stars have in common? They're both sources of heat: the latter a direct one. Let's take that into consideration, and deem this statement true, ignoring any other possible exceptions. Why doesn't a direct source of heat scintillate (given that stars are a main source of heat and that there seems to be a correlation between heat and light for the sake of our theory)? Fire doesn't scintillate, the movements its flames make are of convection- not quite the same, even if an unknowing person could chalk up any star's movements to this. So this theory is once again thrown away. At least from our personal point of view, what we have previously stated is true but not pertinent, thus we take into consideration a new factor that would also assume distance plays a part in the scintillation. And this is a step in the right direction! Let's go back to my first example: Distance makes stars twinkle. This isn't an entirely incorrect statement: distance is a factor that comes into play. In this case, I'm referring to a specific system of measurement. This is called Angular Size. For those of you unaware, angular size is simply the size of

spatial objects observed from the earth. It's measured in degrees, angle, arcseconds and arcminutes. In other words, the distance we view any object in space viewed from earth, measured.
For example, Jupiter is around 30-50 arcseconds. One arcsecond is 1/3600º. Why does this matter, if it's not the fully correct answer to the question? Distance does *in part* change the effects of scintillation on earth, as we were able to identify Jupiter for its lack of twinkling. But the reason stars twinkle is because of a much less intuitive answer, but with logic can be reached. Let us go to the moon, for a moment, or any planet without any atmosphere. A few of you might have already realised where I'm getting at. On the moon, if we were to look out into space, every star wouldn't scintillate. It would seem more akin to a fixed spot in space (not considering the fact that stars have their own orbits and movements). Our conclusion, then, is this: The light that emanates from a star travels through space and reaches our atmosphere, where it is then distorted. On planets where the atmosphere is thin or nonexistent, the stars don't scintillate at all. Angular size is fundamental because the further away from the earth or any planet with a similar type of atmosphere, the more it scintillates, because the light must travel through further distances to reach our earth.